

Expt 5	Design of High Pass Filter using Windowing Techniques
Date:	5.3. Generate the coefficients of an FIR High-Pass filter using Hanning windowing techniques

CODE :-

```

clc; clear; close all;

%% --- Input Parameters ---

fp = 1000; % High-pass cutoff frequency (Hz)
fs = 8000; % Sampling frequency (Hz)
N = 50; % Filter order

wc = fp/(fs/2); % Normalized cutoff (0–1)

%% --- FIR High Pass using Hanning Window ---

b_hann = fir1(N, wc, 'high', hanning(N+1));
disp('--- High-Pass Coefficients (Hanning Window) ---');
disp(b_hann);

%% --- Frequency Response Plots ---

figure;
freqz(b_hann,1);
title('High-pass FIR using Hanning Window');

```

OUTPUT :-

--- High-Pass Coefficients (Hanning Window) ---

-0.0000

0

0.0003

0.0008

0.0009

0
-0.0020
-0.0038
-0.0035
0
0.0057
0.0100
0.0087
0
-0.0127
-0.0216
-0.0183
0
0.0267
0.0464
0.0410
0
-0.0726
-0.1568
-0.2243
0.7500
-0.2243
-0.1568
-0.0726
0
0.0410
0.0464
0.0267
0

-0.0183

-0.0216

-0.0127

0

0.0087

0.0100

0.0057

0

-0.0035

-0.0038

-0.0020

0

0.0009

0.0008

0.0003

0

-0.0000

